D3. Categorical outcomes: tables, tests and effect measures

Binary or nominal outcome compared between groups or exposures. Always state the denominator.

Categorical outcome by group or exposure
e.g. high BP (yes/no) by smoking; adverse event by programme

Start with a contingency table: counts and row percentages
Row percentage = number with the outcome / row total (e.g. smokers with high BP / all smokers)

```
tab <- table(hearts$smoker, hearts$high_bp_0)
prop.table(tab, margin = 1)
```

Are the observations independent or paired?

- independent → **Independent groups**
  Are the expected counts large enough (about 5 or more in each cell)?
  - yes → **Chi-square test**
    ```
    chisq.test(tab)
    chisq.test(tab)$expected
    ```
    2 x 2 tables: Yates' correction by default. Works for larger r x c tables too.
  - no, small → **Fisher's exact test**
    ```
    fisher.test(tab_ae)
    ```
    No large-sample approximation; usual choice when expected counts are small (rare outcomes, small groups).
- paired → **McNemar's test**
  Same people, yes/no twice (high BP at baseline and 12 months), or matched pairs. Only pairs that changed carry information.
  ```
  mcnemar.test(tab_paired)
  ```

Ordinal outcome (e.g. satisfaction 1 to 5): the chi-square test ignores the order. Consider a rank-based test (D2) or ordinal regression (further learning).

A significant test says the proportions differ somewhere. In tables larger than 2 x 2, look at the row percentages to see where. Report the effect measure (right) with its 95% CI, not only the p-value.

Which effect measure? It depends on the study design

Cohort study or randomised trial
People followed over time, so proportions are **risks**. Report risk difference (RD) and risk ratio (RR); OR also possible.
Trial example, adverse events: 14/240 (5.8%) with the programme vs 3/240 (1.2%) with standard advice. RR 4.67 (95% CI 1.36 to 16.03).

Cross-sectional study (one time point)
Proportions are **prevalences**. Report prevalence difference and prevalence ratio (PR); OR also possible.
Course example, high BP: 38/79 (48%) smokers vs 86/401 (21%) non-smokers. PR 2.24 (1.67 to 3.02); OR 3.39 (2.06 to 5.61).

Case-control study
Cases and controls are chosen by outcome, so the proportion with the outcome is set by the design. Risks cannot be estimated: **only the OR** is interpretable.

OR is close to RR only when the outcome is rare (roughly under 10%). With a common outcome the OR is further from 1: do not read an OR of 3.4 as "3.4 times the risk".

Adjusting for a confounder
Stratify and combine: `mantelhaen.test(...)`, or logistic regression for an adjusted OR (aOR): see D4.